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RESEARCH INTERESTS

Data-efficient geospatial machine learning for climate and agricultural risk in places where ground truth is scarce. I work on evaluation and audit methodology for satellite and drone remote sensing, calibrated uncertainty for decision support, and transfer across regions. Nepal's Terai is my main test bed.

EDUCATION

Knox College

September 2023 to June 2027 (expected)

Bachelor of Science, Computer Science and Data Science (double major); GPA 3.82; Dean's List

Galesburg, Illinois

- **Relevant coursework:** Machine Learning; Artificial Intelligence; Models and Complexity; Algorithm Design and Analysis; Statistics and Probability; Linear Algebra; Differential Equations; Research Seminar and Project

PUBLICATIONS

Accepted

- [1] Santosh Pant. *Auditing GeoFM Evaluation for Field-Extent Segmentation: Label Proxies, Baselines, and When Frozen Features Match Fine-tuning*. Transactions on Machine Learning Research (TMLR), 2026. Single author. Accepted August 2026. openreview.net/forum?id=qRXVTe1yYp; [code: github.com/thesantoshpant/auditing-geofm-evaluation](https://github.com/thesantoshpant/auditing-geofm-evaluation); [artifact: doi:10.5281/zenodo.21993508](https://zenodo.org/record/21993508).
- [2] Santosh Pant. *Calibrated District-Level Prediction Intervals for 2025 Paddy Yield Departure in Nepal's Madhesh Province*. Proceedings of the 34th ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems (SIGSPATIAL 2026), Applications track (4-page paper), Riverside, California, November 2026. Single author. [doi:10.1145/3841645.3843341](https://zenodo.org/record/3841645.3843341); [code and provenance: doi:10.5281/zenodo.22184073](https://zenodo.org/record/22184073).

Under review

- [3] Santosh Pant, Adithya Vaithinathan Asokan, Haoyun Feng, Matthew R. Dennis, Keenan Gibbons, Verrah Akinyi Otiende, and Geoffrey Siwo. *Auditing Split Proximity and Radiometric Scale in RGB-to-Thermal Evaluation*. Submitted to the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV 2027), Evaluations and Datasets track, August 2026. Under review.
- [4] Santosh Pant, Chau Tran, Dipsha Budhathoki, and Arito Nakamichi. *Crop-Label Alignment: Auditing Crop-Dependent Auto-Labels in Remote-Sensing Segmentation*. Submitted to the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV 2027), Evaluations and Datasets track, August 2026. Under review.
- [5] Santosh Pant. *When Should a Local Model Correct a Global Cropland Map?*. Submitted to the Tackling Climate Change with Machine Learning workshop at NeurIPS 2026, August 2026. Single author. Under review.

RESEARCH EXPERIENCE

Research Collaborator, RGB-to-Thermal Urban Heat Mapping

March to August 2026

University of Michigan Center for Global Health Equity; Mentor: Prof. Geoffrey H. Siwo

Remote

- Continued the second-place UMich heat-resilience hackathon project with the organizing professor. Diagnosed two silent data faults (colormapped JPEG values used as thermal targets, and RGB and thermal frames misaligned by field-of-view mismatch) and rebuilt the pipeline.
- Developed two training-free checks for RGB-to-thermal benchmarks. The Proximity Audit exposes train-test acquisition-group overlap (on CART, 97.8 to 100 percent of mixing-protocol test frames share a flight with training frames) and the Affine Probe shows that per-frame normalization erases the physical temperature scale before scoring. Led to [3].

Crop-Label Alignment Audit for Remote-Sensing Segmentation

March to August 2026

Knox College; Advisor: Prof. Jurdana Masuma Iqrah

Galesburg, Illinois

- Grew out of a Research Seminar project fusing Sentinel-2 imagery with ICESat-2 altimetry for Antarctic sea-ice classification (model implementation, with Chau Tran, Dipsha Budhathoki, and Arito Nakamichi).
- Introduced crop-label alignment, a diagnostic for pipelines that fit label rules inside each image crop; it is identically zero for any crop-invariant predictor. Calibrated it on Sen1Floods11 and applied it to a 17-acquisition sea-ice benchmark, where the model trained on per-patch labels shows a positive alignment contrast against a scene-label control in all 17 folds. Led to [4].

Sparse-Label Cropland Mapping, Nepal Terai*Independent research, Knox College***2026***Galesburg, Illinois*

- Replicated the Kerner et al. (Tackling Climate Change with ML workshop, NeurIPS 2020) multi-headed LSTM cropland pipeline on Sentinel-2 time series, extended it to Nepal's Terai with sparse CropHarvest labels, and documented a label-source confound in the public replication data.
- Reframed the question as when a local model should be allowed to correct a global cropland map (ESA WorldCereal). Derived metric-specific break-even conditions for map edits, showed that conclusions flip between F0.5 and F2, and found that blind edits move F1 in both directions across seven datasets. Proposed a stratified label screen before editing. Led to [5].

Conformal Prediction Intervals for District Paddy-Yield Loss, Madhesh Province*Independent research, Knox College***2026***Galesburg, Illinois*

- Built leak-controlled district-level prediction intervals for Nepal's 2025 Madhesh drought from a deterministic five-model LightGBM ensemble with split and covariate-shift-weighted conformal calibration on Earth Engine features, plus a Bihar transfer diagnostic; the 2025 decision output was 0 screen-positive, 8 inconclusive, and 0 infinite intervals across eight districts. Led to [2].

Geospatial Foundation-Model Evaluation Audit*Independent research, Knox College***2025 to 2026***Galesburg, Illinois*

- Audited how label proxies and baselines change geospatial foundation-model rankings for field-extent segmentation on single-date Sentinel-2 imagery from six countries (Fields of The World). Replacing field polygons with ESA WorldCover cropland raises random-forest AUROC by up to 0.34, and a from-scratch U-Net beats a Prithvi frozen probe in five of six regions.
- Compared frozen-backbone decoders with full fine-tuning for Prithvi-EO-2.0 and TerraMind over ten matched seeds. The two are TOST-equivalent within 0.02 AUROC in 9 of 12 region-by-model cells, with Kenya (about 0.1 percent positive pixels) the common failure, and the frozen Prithvi decoder trains about 110 times fewer parameters. Four self-audit rounds narrowed the claims before submission. Led to [1].

Satellite Heat-Stress Classification, Nepal Terai*Independent research, Knox College***2026***Galesburg, Illinois*

- Built an early-season vegetation heat-stress classification pipeline over 21 Terai rice districts in Google Earth Engine (MODIS LST and NDVI, ERA5-Land, ESA WorldCover; 59,793 pixel-years) and deployed a live forecast app (terai-heat-forecaster.streamlit.app).
- Under year-blocked leave-one-district-out evaluation with a paired district-block bootstrap, showed that previous-year persistence outperforms a fold-safe XGBoost ensemble on average (macro-F1 0.476 vs 0.362) and proposed persistence and modal-class nulls as required baselines for sub-national benchmarks.

Cropland-Attribution Audit of Satellite Fire Detections, Nepal Terai, Punjab, Madhya Pradesh*Independent research, Knox College***2026 to present***Galesburg, Illinois*

- Auditing how the cropland-restriction step changes MODIS and VIIRS crop-residue burning estimates across regions with very different field sizes, using four independent cropland masks and a published district-level benchmark as a credibility check; manuscript in preparation for a remote-sensing journal.

Richter Research Scholar: AI-Driven Image Analysis of Liquid Crystals*Knox College; Advisor: Prof. Michael Gerten***Summer 2025***Galesburg, Illinois*

- Trained PyTorch convolutional networks for edge detection and phase-transition identification in liquid-crystal microscopy images; applied Grad-CAM to check model attention against domain expectations; evaluated with F1 and Dice under small-sample constraints. Presented as a talk at Knox Horizons, May 2026.

Richter Research Scholar: earlier funded research terms*Knox College; Advisors: Prof. Michael Gerten and Prof. David Bunde***2024 to 2026***Galesburg, Illinois*

- Winter 2025 to 2026 (Gerten): benchmarking and testing tooling for chemical reaction network models (ChemTest).
- Winter 2024 to 2025 (Bunde): simulated 1,000+ configurations of 3,000-node networks to find structured-random-graph properties that limit failure propagation in HPC interconnects; co-authored with Dipesh Adhikari; presented at Knox Horizons, May 2025.
- Summer 2024 (Gerten): built Klicker, a web-based classroom polling and attendance platform, with teammate Ridham Dholaria; presented as a poster at Knox Horizons (May 2025) and, by Ridham Dholaria, at the Midstates Consortium Undergraduate Research Symposium (Washington University in St. Louis, 2025).

PROFESSIONAL EXPERIENCE**Software Engineering Practicum, external client: The MACRO Consortium***Knox College, Software Engineering course***March to June 2025***Galesburg, Illinois*

- Designed and shipped a priority-based scheduling algorithm for the open-source pyscope astronomy package, investigated and mitigated a data-loss bug in its request handling, and added a pytest suite and documentation.

- Built an internal full-stack stock-market simulation platform (React, Express.js, PostgreSQL) for backtesting trading strategies, with real-time ingestion APIs, P&L visualization, Redis caching, and k6 load validation.

SELECTED SOFTWARE PROJECTS

bhasha-js: internationalization platform for South Asian languages

2025 to present

TypeScript, React, Node.js, Express, MongoDB, Gemini (Vertex AI); solo; published on npm

- Built and published an open-source i18n platform: an npm SDK, an Express and MongoDB API, and a React dashboard serving AI-drafted translations across 19 locales, with formal and casual register variants, code-mixed locales such as Hinglish and Roman Nepali, and a compliance lock that keeps regulated strings behind human approval with an atomic audit trail. About 500 automated tests; deployed on GCP, Vercel, and MongoDB Atlas (npmjs.com/package/bhasha-js).

CropScan: calibrated crop-disease diagnosis for small growers

2026

PyTorch (DINOv2 with LoRA, EfficientNetV2-S), FastAPI, React, MongoDB, Docker; three-person team: I built the ML pipeline and most of the backend, teammates did the frontend, design, and documentation

- A leaf-validity gate followed by a calibrated two-model ensemble that commits to a diagnosis only when both models agree and otherwise asks for a better photo. Trained on 69,674 images across 38 diseases and 14 crops; 75.6 percent accuracy on real field photos (PlantDoc) against 99.7 percent on lab images, reported openly. Shaped by 30+ user interviews at Knox Farm; live at cropscan.tech.

Vigilante AI: searchable intelligence from CCTV footage

HackIllinois 2026

FastAPI, Modal (serverless A100), YOLO11, CLIP ViT-L/14, Qwen2.5-VL, Actian VectorAI, React; two-person team

- Runs object detection, CLIP embedding search, and a vision-language model in parallel on one serverless GPU, then serves hybrid semantic search and plain-English chat over the footage with streamed answers and PDF incident reports; won its sponsor track (github.com/thesantoshpant/vigilant-ai).

HONORS AND AWARDS

- **Richter Research Scholar, Knox College:** four funded research terms, 2024 to 2026.
- **Break Through Tech AI Fellow, Cornell Tech:** 2026 to present.
- **HackIllinois 2026 (University of Illinois Urbana-Champaign):** 1st place, Best Use of Actian VectorAI DB (sponsor track), for Vigilante AI.
- **University of Michigan AI for Urban Heat Resilience Hackathon 2026:** 2nd place (\$2,000 prize), for Game of Drones.
- **ASA DataFest 2026 (Central Illinois):** Best Insights Award, team Outliers, for a patient-journey analysis of about 200,000 de-identified health records joined to Census and CDC PLACES data (survival curves, logistic regression, multiple-testing correction).
- **HackAugie 2026 (Augustana College):** Best Gaming Hack.
- **ICPC North Central Regional,** participant (UIUC site), 2025.

TEACHING AND SERVICE

- Teaching Assistant, Data Structures and Algorithms, Knox College (two terms).
- Computer Science Tutor, Knox College, 2023 to present.
- Computer Science Club, lead, Knox College, 2025 to present.

TECHNICAL SKILLS

- **Geospatial and remote sensing:** Google Earth Engine, Rasterio, geopandas, Sentinel-1/2, MODIS, Landsat, ERA5, CHIRPS, ESA WorldCover, ESA WorldCereal, PlanetScope, Fields of The World, CropHarvest, drone RGB and thermal imagery
- **Machine learning:** PyTorch, scikit-learn, XGBoost, LightGBM, EO foundation models (Prithvi-EO-2.0, TerraMind, Clay, AnySat), CNN and U-Net segmentation, LSTM time-series models, Grad-CAM
- **Methods:** split and covariate-shift-weighted conformal prediction, TOST equivalence testing, leakage-aware spatial and temporal evaluation, paired and block bootstrap, multiple-testing correction, seed-matched reporting, preregistration and adversarial self-review
- **Languages and tools:** Python, R, Go, TypeScript, SQL; FastAPI, Streamlit, Folium, React, PostgreSQL, MongoDB, Docker, Git, Linux, uv, pytest